



SFIAD: Deepfake detection through spatial-frequency feature integration and dynamic margin optimization

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Abstract

The rapid advancement of generative models has profoundly transformed the field of digital content creation, bringing unprecedented opportunities for media generation. However, the widespread adoption of this technology has also led to the emergence of highly realistic fake facial images and videos, which pose significant threats to public trust and societal security. To address the challenges of deepfake detection, this paper proposes a novel method based on Spatial-Frequency Feature Integration (SFFI), which effectively identifies fake content by combining spatial and frequency features of images. Additionally, to tackle the issue of class imbalance in the datasets, we propose an Authenticity-Aware Margin Loss (AAML). This loss function dynamically adjusts the decision boundary to enhance the model's ability to recognize minority class samples. The proposed method was trained and evaluated on four challenging datasets: FaceForensics++, Celeb-DF v1, Celeb-DF v2, and the DeepFake Detection Challenge Preview, and compared against ten state-of-the-art methods. Experimental results demonstrate that the proposed method consistently outperforms all existing approaches across all datasets.

Keywords Generative models · Deepfake · Deepfake detection · SFFI · AAML

1 Introduction

Deep generative models (Goodfellow et al. 2020; Karras et al. 2019) have made breakthroughs in digital content creation. They could generate many fake face images and videos (Chan et al. 2019; Gao et al. 2021; Thies et al. 2019) but simultaneously impose a serious challenge to the authenticity of information, especially in terms of privacy invasion and political speech manipulation. To this end, deepfake detection technology has emerged.

Early Deepfake detection methods (Dang et al. 2020; Li et al. 2020; Nguyen et al. 2019; Zi et al. 2020; Afchar et al. 2018; Rossler et al. 2019) primarily relied on convolutional neural networks (CNNs) to extract local facial features. While effective in specific tasks, their

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applicability to Deepfake detection remains limited. These approaches often overlook the complex global information and frequency domain features inherent in Deepfake images, making them inadequate for handling sophisticated forgeries. To address these limitations, recent studies have shifted focus to identifying specific patterns in forged images, such as noise features (Gu et al. 2022; Zhou et al. 2017), local textures (Chen et al. 2021; Gu et al. 2022; Zhao et al. 2021), and frequency information (Li et al. 2021; Qian et al. 2020), to enhance detection accuracy. However, relying on patterns learned from training data renders these methods significantly constrained when facing novel forgery techniques.

Currently, the focus of deepfake detection research has gradually shifted to improving model generalization. For example, Yan et al. (2023) proposed a decoupling framework that enhances generalization by separating forgery traces from content information. However, this method can still capture specific forgery patterns, limiting its adaptability to unknown forgery techniques. Yan et al. (2024) proposed the LSAD method based on a teacher-student architecture, where the teacher module learns domain-specific features through domain loss, and the student module enhances generalization. However, the performance of LSAD is highly dependent on the diversity of the training data, and its generalization ability is limited when data is insufficient. Moreover, these methods operate over the entire feature space, making them vulnerable to irrelevant factors such as background and identity, which can reduce detection accuracy.

To address this issue, this paper proposes a novel detection framework, Spatial-Frequency Feature Integration and Dynamic Margin Optimization (SFIAD), whose overall architecture is illustrated in Fig. 1, consisting of four key components: Spatial-Frequency Feature Integration (SFFI), a pre-trained Xception model, a frame-level classifier, and the Authenticity-Aware Margin Loss (AAML) loss function. The SFFI method combines the original color image with frequency domain information, mitigating cross-image inconsistencies caused by variations in brightness and contrast. The normalization process enhances the model's sensitivity to structural information, significantly improving its robustness against various forgery techniques. We employ the ImageNet-pre-trained Xception network to extract hierarchical features capturing complex image patterns.

The frame-level classifier processes each frame to extract key features, laying the foundation for video content understanding and improved classification accuracy. By dynamically adjusting class-specific decision boundaries according to each class's sample distribution, the AAML loss function increases weight allocation to real images and encourages focused model attention on discriminative features, thus significantly enhancing classification performance, especially on highly imbalanced datasets.

Our contributions are summarized as follows:

- We propose the SFIAD, a novel deepfake detection method that innovatively fuses spatial-frequency features with dynamic boundary optimization, enabling robust identification across unknown data environments.
- We propose the SFFI module, which integrates spatial and frequency domain information from video frames to enhance feature extraction capabilities, simultaneously capturing detailed texture and frequency domain features and improving the model's performance in complex visual tasks.
- This paper proposes the AAML loss, which addresses the class imbalance problem in deepfake detection through dynamic boundary adjustment, enabling the model to focus



Fig. 1 In this study, we compared four classification methods: the traditional approach based on a complete feature set, an enhanced approach leveraging auxiliary information for improved feature representation, a task-specific approach with a customized loss function, and our proposed method, which integrates the strengths of the first three, achieving superior classification performance

more effectively on rare real samples and significantly enhancing real-image recognition accuracy.

2 Related work

2.1 Deepfake generation

Traditional image and video forgery techniques typically involve intercepting source content and embedding target objects, using simple pixel-level operations to create fake images (Wang et al. 2022; Rossler et al. 2019). However, with the rapid advancement of deep learning (Muksimova et al. 2024), deepfake technology has made significant breakthroughs, greatly enhancing the realism of forged content and making it nearly indistinguishable. Early forgery video generation methods relied on autoencoders (D'Avino et al. 2017) for image reconstruction, but these methods were typically limited to simple reconstruction of input data, resulting in lower-quality fake images.

The advent of Generative Adversarial Networks (Karras et al. 2019; Li et al. 2018) has revolutionized the generation of forged content. Through the adversarial training between the generator and the discriminator, GANs can generate fake images or videos that are visually more realistic and rich in detail. In recent years, GAN-based deepfake technology has made significant progress in areas such as face swapping, voice synthesis, and facial expression manipulation. The introduction of Conditional GANs (Odena et al. 2017; Mirza 2014) and Cycle Consistent Networks (Zhu et al. 2017; Wang et al. 2018; Huang et al. 2018) has particularly improved the quality and consistency of generated images and videos.

However, as technology advances, the generation of fake content has become increasingly complex, especially with the emergence of diffusion models (Ho et al. 2020; Saharia et al. 2022). Unlike GANs, diffusion models provide more refined control over the generation process by simulating a step-by-step denoising process, achieving new heights in the detail and consistency of forged content. This technological advancement has further enhanced the realism of fake images and videos, posing new challenges for traditional forgery detection methods. Therefore, in response to the growing complexity of forgery technology, the deepfake detection field urgently requires more powerful and flexible detection methods.

2.2 Deepfake detection

In the field of deep forgery detection, the generalization ability of the model, especially the adaptability to different forgery techniques and datasets, remains a key challenge. Existing research focuses on two main directions: frame-level forgery detection and video-level forgery detection. Frame-level forgery detection focuses on extracting forgery features from a single image frame, while video-level forgery detection identifies forged content by analyzing the temporal relationship between video frames.

In frame-level forgery detection, Chen et al. (2022) proposed a detection method combining self-supervised learning and adversarial data augmentation, effectively enhancing the model's adaptability to unknown forgery techniques by generating diverse forged images. Liu et al. (2021) introduced the SPSL method, which combines spatial domain and phase spectrum information, utilizing shallow networks to focus on local texture features, significantly improving the model's generalization ability across different datasets. Luo et al. (2021) further enhanced the model's cross-database consistency by incorporating high-frequency noise features with attention mechanisms. Dong et al. (2023) proposed a detection method that does not rely on identity information, focusing on local forgery features, thus improving the model's adaptability to unknown forgery techniques. Huang et al. (2023) introduced explicit and implicit identity loss, proposing a framework driven by implicit identities, which effectively enhanced the model's generalization ability.

Additionally, feature disentangling learning and reconstruction-based learning methods have shown significant potential in deepfake detection. Yan et al. (2023) proposed a multi-task disentangling framework combining conditional decoders and contrastive regularization, which effectively extracts common features across different forgery techniques, improving detection accuracy. Liang et al. (2022) introduced a framework that constrains feature independence to suppress the interference of content information, enhancing the identification of forgery traces. In terms of network design, Dang et al. (2020) proposed a novel attention mechanism and demonstrated its effectiveness on large-scale forgery datasets. Zhao

et al. (2021) introduced a multi-attention network that combines region-independent loss and attention-guided data augmentation strategies, achieving state-of-the-art performance across multiple datasets. Cao et al. (2022) proposed an end-to-end reconstruction classification framework (RECCE), which enhances the model's robustness to unknown forgery patterns by reconstructing real images and exploring the inherent differences between real and forged images. Wang and Deng (2021) proposed the Forgery Region Mining method (RFM), which integrates attention-guided data augmentation to improve detection accuracy and can be seamlessly integrated into existing CNN models. Zhu et al. (2021) introduced a 3D decomposition method and designed a dual-stream network architecture (FD2Net), which achieved excellent detection performance across multiple benchmark datasets.

In contrast, video-level forgery detection mainly focuses on the temporal relationship between video frames, especially abnormal temporal consistency, eye blinking, and optical flow features in forged videos. Haliassos et al. (2021) present a spatio-temporal network pre-training method that demonstrates good robustness in response to perturbations by learning natural mouth movement patterns and fine-tuning the network to detect forged videos. Li et al. (2018) used a Long Short-Term Memory convolutional network (LRCN) to analyze the blink pattern and successfully identified the missing physiological signal in the forged video, which significantly improved the detection efficiency. FakeSpotter method proposed by Wang et al. (2019) shows strong anti-perturbation ability by monitoring neuron behavior and using average neuron coverage (ANC) to detect fake faces generated by GAN. Amerini et al. (2019), combined with the inconsistency of optical flow field, proposed a detection method based on dynamic features, which can effectively deal with the inconsistency problem across sequences.

3 Method

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- 1: **Input:** Color image I_{color}
 - 2: **Output:** Classification result
 - 3: **Procedure:**
 - 4: Convert the color image I_{color} to a grayscale image I_{gray}
 - 5: Apply a two-dimensional Fast Fourier Transform (FFT) to I_{gray} to obtain the frequency domain representation $F(I_{\text{gray}})$
 - 6: Center the spectrum using fftshift to get $F_{\text{shift}}(I_{\text{gray}})$
 - 7: Apply a logarithmic transformation to compress the dynamic range and highlight detailed features, resulting in $A_{\log}$
 - 8: Normalize the FFT image to bring values within the range $[0, 1]$ to get A_{norm}
 - 9: Concatenate the original color image I_{color} and the normalized frequency domain features A_{norm} along the channel dimension to form I_{combined}
 - 10: Extract features from I_{combined} using a pre-trained Xception model to get featured
 - 11: Train a classifier using the extracted features and associated labels
 - 12: During training, use Authenticity-Aware Margin Loss (AAML) to calculate the loss L
 - 13: Update the model parameters to minimize L
 - 14: **return** The classification result for the input image
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Algorithm 1 Spatial-Frequency Feature Integration and Classification

This section provides an overview of the intuitive design of our method and details the network architecture for video frame processing based on SFFI. Additionally, we propose a loss function to address the class imbalance issue.

3.1 The intuitive idea behind our method

3.1.1 An intuitive understanding of SFFI

Forgery detection should not be limited to spatial domain analysis of images but should also delve into the frequency domain to reveal subtle features of forged content. Forged images often exhibit unnatural high-frequency noise and abnormal frequency distributions in the frequency domain, which are difficult to capture through traditional spatial domain analysis. Our intuition is that by combining spatial domain features with frequency domain features, we can more comprehensively capture the subtle differences in forged images.

3.1.2 An intuitive understanding of AAML

Class imbalance is a common issue in deep fake detection, where the number of real samples is significantly lower than that of fake samples, leading to a reduced ability of the model to recognize real samples during training. To address this issue, a loss function is proposed that dynamically adjusts the decision boundary based on class imbalance.

3.2 Overview

The proposed network architecture is shown in Fig. 2. To address the widespread issue of facial forgery in current deepfake detection datasets, we propose a comprehensive method, and its data processing and processing workflow are as follows:

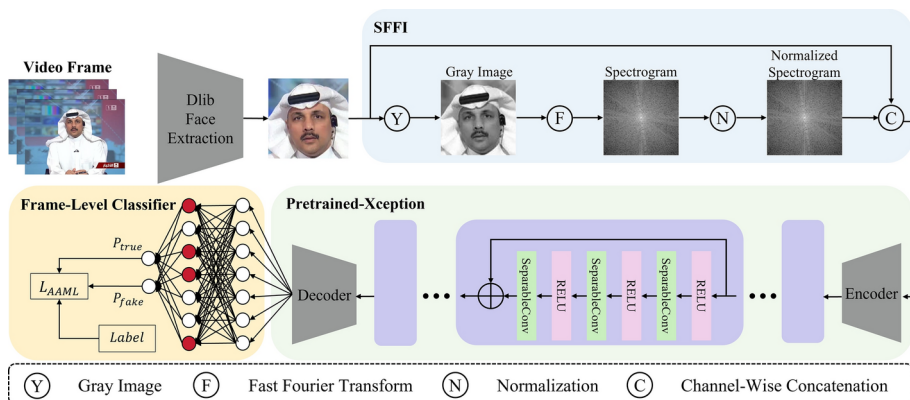


Fig. 2 The architecture of SFIAD. The extracted video frame sequence is first passed through the Dlib face detector for face extraction, which serves as the model input. The SFFI fuses the spatial and frequency domain features of the image, and the Pretrained-Xception model is used for feature extraction. The resulting feature map is then input into the frame-level classifier for classification. Finally, the model output is compared with the ground truth labels to compute the loss

1. **Face region extraction:** The Dlib face detector (Sagonas et al. 2016) performs frame-wise facial region detection on input videos, which constrains model attention to discriminative facial features and enhances computational efficiency by reducing irrelevant background processing.
2. **Pre-trained xception:** The ImageNet-pre-trained Xception backbone effectively captures complex image patterns through its deep convolutional layers. It processes SFFI outputs as a hierarchical feature extractor (with the final FC layer removed) to encode discriminative spatial-spectral representations.
3. **Frame-level classifier:** The extracted feature maps are processed by a lightweight classifier comprising FC layers, which performs frame-wise authenticity classification by leveraging discriminative feature representations while maintaining temporal coherence across video sequences.

To address the class imbalance problem between real and fake samples in deepfake detection, we propose the AAML loss function. This loss function dynamically adjusts the decision boundary based on the sample distribution of each class, assigning a larger margin to real images, thereby increasing their weight during training and encouraging the model to focus more on recognizing real images. This design significantly improves classification performance on imbalanced datasets. To illustrate the steps of the proposed method, Algorithm 1 shows the overall process of our deepfake detection algorithm. The current framework focuses on frame-level detection. When conducting subsequent research on video-level detection, a spatio-temporal feature processing module will be added based on the existing one. This module will be placed between the SFFI module and the pre-trained Xception model. It is responsible for extracting the spatio-temporal features of video frame sequences. For instance, it can use the optical flow method to obtain the speed and direction information of facial movements. Then, these spatio-temporal features will be fused with the single-frame features extracted by the SFFI module and input into the pre-trained Xception model for more advanced feature extraction and analysis, thereby achieving more accurate detection of video-level deepfakes.

3.3 Detailed network architecture

As illustrated in Fig. 2, our network model consists of three main components: an SFFI module, a Pretrained-Xception (Chollet 2017) model, and a Frame-Level Classifier. Figure 2 illustrates the specific implementation of these three modules used in our experimental evaluation. We assume that the input video contains m frames, with each frame having a size of $h \times w \times 3$. Therefore, the input size to our model is $m \times h \times w \times 3$.

3.3.1 SFFI

Previous research has mainly focused on spatial domain features of images, overlooking the importance of frequency domain information. Furthermore, there has been no effective method to integrate spatial and frequency domain information. As shown in Fig. 3, significant differences exist between real and forged images in the frequency domain. Based on this observation, this paper proposes the SFFI method, which combines the original color

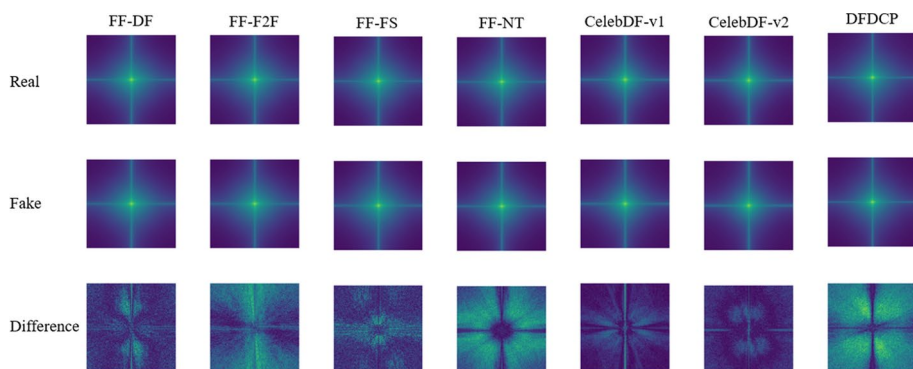


Fig. 3 Frequency analysis for each dataset. We present the average spectra of images, focusing on real and fake images. We observe that artifacts are evident in the average spectra of the displayed deepfake forgeries

image with frequency domain information, enabling the model to capture more useful features during the feature extraction phase and enhancing its recognition ability.

To be specific, we first extract the color image $I_{\text{color}} \in \mathbb{R}^{H \times W \times C}$ from the input data dictionary and convert it to a grayscale image by averaging across channels to obtain a single-channel grayscale image $I_{\text{gray}} \in \mathbb{R}^{H \times W \times 1}$:

$$I_{\text{gray}}(x, y) = \frac{1}{C} \sum_{c=1}^C I_{\text{color}}(x, y, c) \quad (1)$$

where $I_{\text{gray}}(x, y)$ is the pixel value of the grayscale image, C is the number of color channels, and x, y are the spatial coordinates of the image. Next, we apply a two-dimensional Fast Fourier Transform (FFT) to the grayscale image to obtain its frequency domain representation $\mathcal{F}(I_{\text{gray}})$:

$$F(I_{\text{gray}})(u, v) = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} I_{\text{gray}}(x, y) \cdot e^{-j2\pi(\frac{ux}{H} + \frac{vy}{W})} \quad (2)$$

where $F(I_{\text{gray}})(u, v)$ is the complex value in the frequency domain, u, v are frequency coordinates, H, W are the height and width of the image, and j is the imaginary unit. And re-arrange the spectrum using centralization (FFTshift):

$$u' = (u + \frac{H}{2}) \bmod H, \quad v' = (v + \frac{W}{2}) \bmod W \quad (3)$$

$$F_{\text{shift}}(I_{\text{gray}})(u, v) = F(I_{\text{gray}})(u', v') \quad (4)$$

where u', v' are the frequency coordinates after centralization. Additionally, to compress the dynamic range of the frequency amplitude, we apply a logarithmic transformation:

$$A_{\log}(u, v) = \log(A(u, v) + \epsilon) \quad (5)$$

where $A(u, v)$ represents the amplitude spectrum and ϵ is a small positive constant used to avoid division by zero in the logarithmic operation, thereby reducing the amplitude differences and highlighting detailed features in the image. After processing the frequency domain, we normalize the FFT image so that its values fall within the range $[0, 1]$:

$$A_{\text{norm}}(u, v) = \frac{A_{\log}(u, v)}{\max(A_{\log}(u, v))} \quad (6)$$

where $\max(A_{\log}(u, v))$ is the maximum amplitude in the spectrum, used for normalization. Finally, we concatenate the normalized frequency domain features with the original color image along the channel dimension, forming a multi-channel representation that incorporates both spatial and frequency domain information:

$$I_{\text{combined}} = \text{Concat}(I_{\text{color}}, A_{\text{norm}}) \in \mathbb{R}^{H \times W \times (C+1)} \quad (7)$$

where I_{combined} is the concatenated image that incorporates both spatial and frequency domain information.

The core idea of this method is to combine the spatial and frequency domain features of video frames, fully leveraging the advantages of the Fast Fourier Transform in capturing frequency information. This allows the model to not only capture detailed texture information of the image but also integrate additional frequency domain features, thereby improving the model's adaptability and accuracy in complex visual tasks. To validate the SFFI module's efficacy, Fig. 4 compares feature activation patterns. Column 1 displays Dlib-extracted facial regions; Column 2 contrasts Xception activation heatmaps between baseline (Row 1: raw inputs) and SFFI-enhanced (Row 2: spectral-processed inputs). Quantitative analysis demonstrates that the SFFI module enhances discriminative feature extraction in high-frequency components (e.g., texture edges and spectral anomalies) while expanding the receptive field. This enables the model to extract information from a broader range of image areas, thereby notably improving forgery detection accuracy through comprehensive spatial-spectral pattern learning.

3.3.2 Pretrained-xception

We utilize a pre-trained convolutional neural network (CNN) module to extract spatial features from each frame. The Xception model, pre-trained on ImageNet, is selected, with its final fully connected (FC) layer removed to serve as a feature extractor. To accommodate a different number of input channels, we modify the weight-loading process of Xception to handle an input format distinct from the original model. After processing, the output feature map from the pre-trained Xception model has a size of $m \times 2048$, where m represents the number of frames and 2048 is the feature dimension.

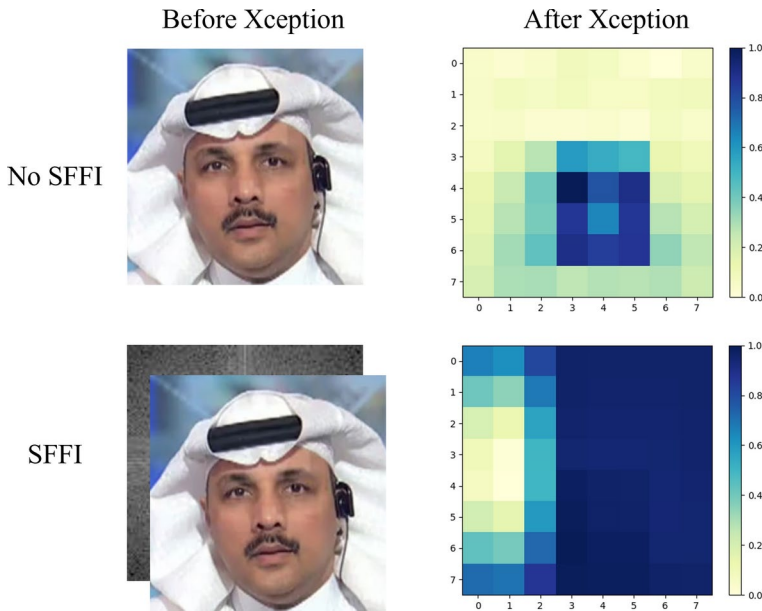


Fig. 4 Visualization of feature maps of the Xception model before and after applying the SFFI module. The receptive field increases after application

3.3.3 Frame-level classifier

The features from the Pretrained-Xception model are fed into the frame-level classifier. It receives feature maps of size 8 for each frame and classifies each frame through a fully connected (FC) layer. In our implementation, the length of the input video is set to 32 frames (more details provided in Sect. 4), and the size of the input frame is 256×256 .

3.4 AAML

In the task of deepfake detection, class imbalance is a critical issue. With fake samples dominating the dataset and authentic samples being relatively scarce, this imbalance weakens the model's ability to recognize the minority class (authentic samples). To address this challenge, we propose the AAML loss function. By dynamically adjusting classification boundaries, AAML significantly enhances the model's ability to distinguish authentic images, particularly in scenarios with severe data imbalance.

The core mechanism of AAML is to dynamically adjust the classification margin based on the sample count of each class, effectively mitigating the impact of class imbalance on model training. Specifically, for the class with fewer samples (authentic samples), AAML assigns a larger margin m_y , while for the class with more samples (fake samples), it sets a smaller margin m_y . This differential margin design encourages the model to focus more on the characteristics of authentic images during classification. Given the sample count n_y for class y , the margin m_y is calculated as follows:

$$m_y = \frac{1}{\sqrt{\sqrt{n_y}}} \cdot \frac{m_{\max}}{\max\left(\frac{1}{\sqrt{\sqrt{n_y}}}\right)} \quad (8)$$

where $m_{\max}$ is the preset maximum margin, and n_y represents the number of samples in class y . With this design, the margin is adaptively scaled based on class frequency, giving the less frequent class (real images) a larger margin, thus increasing its importance during training. The specific results of the marginal calculation are shown in Fig. 5.

Once the margins are defined, we adjust the logits for the model's outputs. For a given input x and target label y , the logits are adjusted by subtracting the corresponding margin m_y as follows:

$$x_{i,y} = x_{i,y} - m_y \quad (9)$$

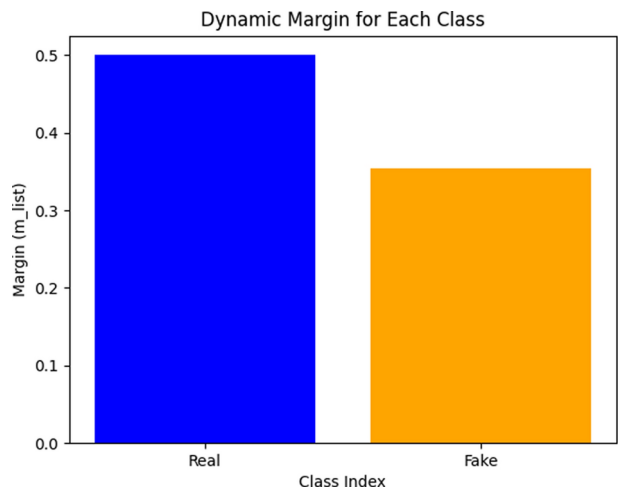
where $x_{i,y}$ is the logit of the i -th input belonging to class y , and m_y is the margin associated with that class. This adjustment allows the model to maintain a clearer distinction between classes, especially in imbalanced datasets.

AAML is based on the standard cross-entropy loss but introduces class-specific margin adjustments to the logits. For a given set of input logits x and the corresponding target labels y , the AAML loss is defined as:

$$L(x, y) = \frac{1}{N} \sum_{i=1}^N \text{CrossEntropy}(s \cdot (x_{i,y} - m_{y_i}), y_i) \quad (10)$$

where s is the scaling factor, and m_y is the margin associated with class y , dynamically adjusted based on the sample count in each class. The cross-entropy loss function is employed to compute the loss on the logits following margin adjustment for each sample. By implementing this approach, AAML establishes a more pronounced distinction constraint on the real image class, effectively mitigating the risk of it being "overwhelmed" by the larger volume of fake images.

Fig. 5 The figure illustrates the dynamic margin adjustment for different classes in the AAML method. For the real image class (Real), a larger margin is assigned due to fewer samples, while a smaller margin is assigned to the fake image class (Fake) because it has more samples. This adjustment helps alleviate the class imbalance issue and enhances the model's ability to recognize real images



By introducing class-specific margins, AAML effectively mitigates the negative impact of class imbalance on model training. This is particularly important in deepfake detection tasks, where real samples are often the minority class. The dynamic margin adjustment mechanism encourages the model to focus more on the minority class, resulting in significantly improved classification accuracy. Furthermore, the margin adjustment process can be seamlessly integrated into existing training workflows with minimal computational overhead, leading to substantial performance improvements.

4 Experiments

4.1 Settings

Datasets Our method has been evaluated on four challenging datasets, including FaceForensics++ (FF++) (Rossler et al. 2019), CelebDF-v1 (CDFv1) (Li et al. 2020), CelebDF-v2 (CDFv2) (Li et al. 2020), and DeepFake Detection Challenge Preview (DFDC-P) (Dolhansky et al. 2019). FF++ is currently the most widely used dataset for synthetic video forgery, consisting of 1000 authentic videos and corresponding manipulated videos generated through four distinct manipulation techniques: Deepfakes (DF), Face2Face (F2F), FaceSwap (FS), and Neural Textures (NT). The FF++ dataset also offers three versions based on compression levels: original, light compression (c23), and heavy compression (c40). Following the convention of previous studies, we have selected the light compression version (c23) of the FF++ dataset for our experiments. This dataset is primarily utilized for model training, with the others reserved for testing purposes.

Evaluation metrics Consistent with prior studies (Yan et al. 2023, 2024), our evaluation employs established metrics: Accuracy (ACC), Average Precision (AP), and the Area Under Curve (AUC), while incorporating F1-score and computational efficiency (GFLOPS/Params) for comprehensive performance characterization.

Implementation details The experiment configuration parameters are given in Table 1. All methods use the same training configuration, including image size, optimizer choice, learning rate, and data augmentation strategy. During the data processing phase, we selected 32 frames from each video sample for training and testing. Additionally, we efficiently executed facial detection, cropping, and alignment tasks using the DLIB library, standardizing the image dimensions to 256×256 pixels. In the training process, we employed the Adam optimization algorithm with a learning rate of 0.0002 and uniformly adopted a batch size of 32. To enhance the robustness of the model, we implemented a range of data augmentation strategies during the training phase, including image compression, horizontal flipping, rotation, Gaussian blurring, and random adjustments to brightness and contrast. All experiments were conducted on NVIDIA GTX 3090 GPUs, ensuring ample computational resources and consistency in the experimental environment. To ensure a fair comparison, all experiments are conducted within the DeepfakeBench (Yan et al. 2023). All our experimental settings follow the default settings of the benchmarks.

Table 1 Experimental setup

Training config	All methods
Image size	256,256
Weight initialization	ImageNet Pre-trained
Optimizer	Adam
Base learning rate	2×10^{-4}
Weight decay	5×10^{-4}
Optimizer momentum	$B1, B2 = 0.9, 0.999$
Batch size	32
Training epochs	10
Learning rate schedule	None, Constant
Flip probability	0.5
Rotate probability	0.5
Rotate limit	$[-10, 10]$
Blur probability	0.5
Blur limit	$[3, 7]$
Brightness probability	0.5
Brightness limit	$[-0.1, 0.1]$
Contrast limit	$[-0.1, 0.1]$
Quality lower	40
Quality upper	100

4.2 Generalization ability evaluation

4.2.1 Comparison with competing methods

To evaluate generalization capability, we benchmarked against ten state-of-the-art methods under consistent experimental settings: MesoNet (Afchar et al. 2018), FWA (Li and Lyu 2018), Xception (Rossler et al. 2019), F³-Net (Qian et al. 2020), RFM (Wang and Deng 2021), SRM (Luo et al. 2021), CORE (Ni et al. 2022), Recce (Cao et al. 2022), UCF (Yan et al. 2023), and LSDA (Yan et al. 2024).

We trained our model on the FF++ dataset and conducted rigorous cross-dataset evaluation on Celeb-DF-V1, Celeb-DF-V2, and DFDC benchmarks. This protocol challenges generalization capability by introducing significant distribution shifts between training (synthetic forgeries) and testing data (realistic deepfakes with diverse generation pipelines), thereby quantitatively assessing model adaptability and cross-dataset robustness through controlled out-of-domain experimentation.

Tables 2 and 3 illustrate comparative results using different methods. We know that SFIAD outperforms other methods across multiple datasets. Specifically, it exhibits remarkable performance in the Celeb-DF-V1 dataset, attaining the highest scores in all evaluation metrics. Notably, its F1 score exceeds that of the second-best method by 4.49%, which achieves an excellent balance between precision and recall through spectral-spatial fusion. The model maintains generalization prowess on Celeb-DF-V2 (achieving the highest scores in ACC, AUC, AP, and F1) through multi-scale feature learning, while sustaining competitive AUC (75.88%) and AP (83.86%) on DFDC despite slight accuracy variance. Furthermore, Table 2 shows an evaluation of computational efficiency in terms of GFLOPS and the number of parameters. Although SFIAD is not the most efficient in these metrics, its

Table 2 Testing on the Celeb-DF-V1 dataset

Methods	GFLOPS	PARAMS	ACC(%)	AUC(%)	AP(%)	F1(%)
MesoNet	2.0159	0.0279	61.64	58.63	64.11	76.27
FWA	192.3708	20.8111	64.06	66.50	70.13	80.60
Xception	192.3708	20.8111	68.59	74.18	82.44	74.74
F ³ -Net	193.7086	20.8136	61.19	62.69	73.32	73.92
RFM	192.3708	20.8111	74.68	80.93	86.38	77.27
SRM	441.9139	53.2384	60.43	60.46	71.02	71.33
CORE	192.3708	20.8111	59.76	62.09	71.98	71.64
Recce	258.2194	23.7836	61.73	62.05	74.46	74.62
UCF	390.1280	44.5052	<u>71.21</u>	76.79	84.10	76.49
LSDA	<u>4.7333</u>	<u>1.3069</u>	68.88	<u>82.46</u>	<u>87.02</u>	<u>79.43</u>
SFIAD	192.5195	20.8113	78.06	88.40	92.45	83.92

Bold and underlined indicate the best and second-best performance, respectively

Table 3 Testing results on Celeb-DF-V2 and DFDCP datasets

Methods	Celeb-DF-V2				DFDCP			
	ACC(%)	AUC(%)	AP(%)	F1(%)	ACC(%)	AUC(%)	AP(%)	F1(%)
MesoNet	64.33	56.43	69.63	77.08	66.10	59.71	70.26	<u>79.49</u>
FWA	65.62	57.96	71.00	61.57	62.51	60.44	73.89	68.96
Xception	60.39	73.21	81.80	62.90	63.54	73.65	82.05	67.49
F ³ -Net	62.03	64.86	77.81	70.08	70.85	73.10	81.07	77.95
RFM	59.84	74.13	83.50	78.71	64.13	72.32	80.30	73.85
SRM	63.67	64.20	76.32	72.72	67.23	73.39	82.16	72.89
CORE	65.43	65.77	77.85	74.98	70.03	74.83	83.16	76.13
Recce	67.47	66.13	77.73	78.30	<u>71.02</u>	74.15	82.09	77.89
UCF	64.32	73.06	81.61	68.97	60.92	75.05	83.00	62.91
LSDA	<u>69.50</u>	<u>75.90</u>	<u>83.44</u>	<u>81.10</u>	67.16	79.68	87.05	79.95
SFIAD	75.19	79.00	86.58	81.88	71.43	<u>75.88</u>	<u>83.96</u>	77.72

Bold and underlined indicate the best and second-best performance, respectively

significant advantage in detection accuracy makes it a practical choice for applications that prioritize high detection performance.

LSDA (Yan et al. 2024) adopts a teacher-student architecture, where the teacher module learns domain-specific features through domain loss, while the student network enhances generalization through feature distillation. However, its performance heavily depends on the diversity of training data. When data is insufficient, LSDA struggles to capture the distribution of forgery features effectively, limiting its adaptability to unknown forgery types. Furthermore, LSDA involves complex latent space operations, has a long training time, and is highly sensitive to hyperparameters, increasing training difficulty and uncertainty.

UCF (Yan et al. 2023) uses a multi-task learning strategy and conditional decoder to decouple image features, extracting general forgery features to improve detection generalization. However, UCF has two main issues: first, the computational complexity of feature decoupling is high, especially on large-scale datasets; second, the decision process relies on a complex decoupling and contrastive learning mechanism, resulting in poor interpretability and making it difficult to explain the detection rationale to users.

SRM (Luo et al. 2021) and F³-Net (Qian et al. 2020) methods distinguish original images from forged ones by leveraging image frequency components and perform reasonably well when dealing with known forgery patterns. However, these two methods heavily rely on the forgery features learned from the training samples. Due to their inability to extract new features, their performance significantly declines when encountering unknown forgery patterns. In comparison, our proposed SFIAD model has significant advantages. First, SFIAD fuses spatial and frequency domain features, broadening the capture range of deepfake features and effectively improving detection capabilities. Second, SFIAD employs Fast Fourier Transform (FFT) to perform spatial spectrum synthesis analysis, overcoming the limitations of traditional methods that rely solely on high-pass filtering or specific frequency features.

CORE (Ni et al. 2022) and Recce (Cao et al. 2022) focus on different detection algorithms, including designing specific loss functions and employing reconstruction learning. However, these detectors can still be affected by irrelevant factors such as race, gender, or identity, making it challenging to eliminate these confounding influences.

FWA (Li and Lyu 2018) method detects fake videos by identifying resolution inconsistencies in face transformations. However, it has two notable limitations: first, it lacks the ability to capture irrelevant transformation artifacts, which leads to a decline in detection performance; second, it relies on specific image processing techniques to generate negative samples, making it prone to overfitting to certain forgery patterns and resulting in poor generalization.

RFM (Wang and Deng 2021) method guides the detector to focus on more representative forged regions through forged attention maps and a suspicious forgery elimination module, improving detection capability across various forgery techniques. However, RFM has two limitations: first, it heavily relies on the quality of training data, with noise or mislabeling significantly affecting model performance; second, its attention mechanism is prone to interference from specific forgery patterns, leading to an overemphasis on local features and reducing the model's generalization ability.

Finally, MesoNet (Afchar et al. 2018) and Xception (Rossler et al. 2019) are pure convolutional network structures that lack data augmentation, feature disentanglement, or utilization of frequency information. Their performance significantly declines when dealing with unknown forged videos.

Figures 6, 7, and 8 show the comparison results of AUC, AP curves, and F1-score bar charts for different methods on the Celeb-DF-v1, Celeb-DF-v2, and DFDCP datasets, respectively. Each line represents the performance of a different model, while the F1-scores are displayed as bar charts. Overall, the SFIAD model performs well on multiple datasets, demonstrating its robustness in handling unknown forgery patterns.

To further verify the effectiveness of our proposed SFIAD method, we conducted a significance analysis comparing its performance with other state-of-the-art methods. Table

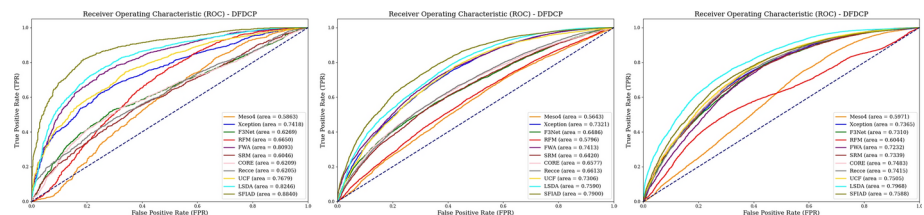


Fig. 6 Cross-dataset evaluation: Plotting the ROC-AUC curve with the optimal model parameters

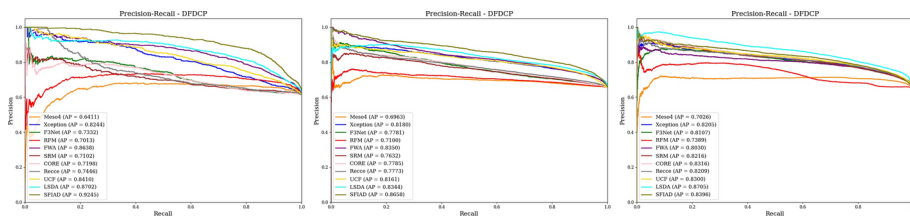


Fig. 7 Cross-dataset evaluation: Plotting the Precision-Recall curve with the optimal model parameters

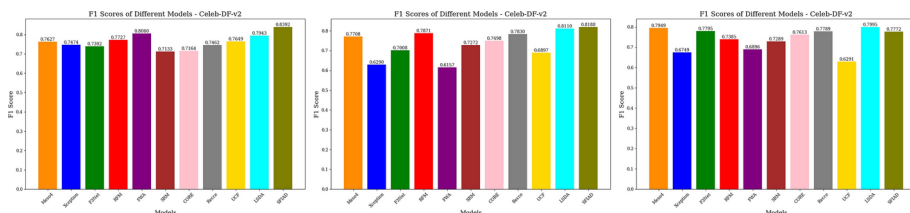


Fig. 8 Cross-dataset evaluation: Plotting the bar chart of F1-score with the optimal model parameters

Table 4 Significance analysis results

Comparison of methods	P value	Level of significance
SFIAD vs. MesoNet	0.009	Significant
SFIAD vs. FWA	0.007	Significant
SFIAD vs. Xception	0.024	Significant
SFIAD vs. F ³ -Net	0.049	Significant
SFIAD vs. RFM	0.137	Not significant
SFIAD vs. SRM	0.015	Significant
SFIAD vs. CORE	0.049	Significant
SFIAD vs. Recce	0.070	Not significant
SFIAD vs. UCF	0.058	Not significant
SFIAD vs. LSDA	0.035	Significant

4 shows the P-values and levels of significance for the performance comparison between SFIAD and other methods.

The significance analysis shows that SFIAD is statistically superior to MesoNet, FWA, Xception, F³-Net, SRM, CORE, and LSDA. However, the performance differences between SFIAD and RFM, Recce, and UCF are not statistically significant. These findings indicate that while SFIAD generally achieves overall performance improvements, further optimization is needed - especially in reducing the false positive rate, where it is comparable to methods such as RFM, Recce, and UCF.

4.3 Ablation study

We conducted ablation experiments to verify the effectiveness of the modules in this paper. The model is trained on the FF++ dataset and evaluated on the other three datasets. This experimental setup demonstrates the challenging nature of models in the evaluation of generalization ability.

To evaluate the impact of the SFFI and AAML modules on the SFIAD method, we performed ablation experiments by removing each module individually as well as both modules simultaneously. The results are summarized in Table 5, which reports accuracy (ACC), area under the curve (AUC), and average precision (AP) on the Celeb-DF-V1, Celeb-DF-V2, and DFDCP datasets.

Removing the SFFI module caused significant performance degradation: on Celeb-DF-V1, ACC decreased from 78.06% to 69.71%, AUC from 88.40% to 81.56%, and AP from 92.45% to 86.86%; on Celeb-DF-V2, ACC decreased from 75.19% to 71.36%, AUC from 79.00% to 76.45%, and AP from 86.58% to 83.29%; on DFDCP, ACC decreased from 71.43% to 66.51%, AUC from 75.88% to 74.82%, and AP from 83.96% to 83.23%.

Similarly, removing the AAML module also led to performance degradation: on Celeb-DF-V1, ACC decreased from 78.06% to 73.28%, AUC from 88.40% to 80.95%, and AP from 92.45% to 87.78%; on Celeb-DF-V2, ACC decreased from 75.19% to 64.35%, AUC from 79.00% to 74.85%, and AP from 86.58% to 84.55%; on DFDCP, ACC decreased from 71.43% to 70.26%, AUC from 75.88% to 74.13%, and AP from 83.96% to 82.13%.

When both modules were removed, the performance decline became even more pronounced: on Celeb-DF-V1, ACC dropped to 68.59%, AUC to 74.18%, and AP to 82.44%; on Celeb-DF-V2, ACC dropped to 60.39%, AUC to 73.21%, and AP to 81.80%; on DFDCP, ACC dropped to 63.54%, AUC to 73.65%, and AP to 82.05%.

These results demonstrate that both the SFFI and AAML modules are crucial for the SFIAD method, as they complement each other to improve the model's ability to distinguish between real and fake data.

In addition, we experimentally verify the effectiveness of AAML. As shown in the Table 6, When the loss function we designed is applied to other methods, all methods improve except Meso4, which has a decrease in DFDCP data.

Table 7 compares the AUC performance of the Xception model combined with different loss functions on the Celeb-DF-V1, Celeb-DF-V2, and DFDCP datasets. The results show that Xception with the AAML loss achieved the highest AUC scores on all three datasets, significantly outperforming Xception models combined with cross-entropy (Krizhevsky et al. 2012), PaCoLoss (Zhu et al. 2022), and AM-Softmax (Wang et al. 2018). This demonstrates the effectiveness of the AAML loss in improving detection performance.

Figure 9 illustrates the feature space distribution based on t-SNE, aiming to analyze the impact of the AAML module on classification boundaries. On the Celeb-DF-V1 test set, the SFIAD model with AAML (SFIAD On Celeb-DF-V1 (test)) significantly improved the separability between real and forged samples, resulting in more discrete feature distribu-

Table 5 Ablation on the network architecture

Methods	Celeb-DF-V1			Celeb-DF-v2			DFDCP		
	ACC(%)	AUC(%)	AP(%)	ACC(%)	AUC(%)	AP(%)	ACC(%)	AUC(%)	AP(%)
SFIAD	78.06	88.40	92.45	75.19	79.00	86.58	71.43	75.88	83.96
SFIAD Del SFFI	69.71	81.56	86.86	71.36	76.45	83.29	66.51	74.82	83.23
SFIAD Del AAML	73.28	80.95	87.78	64.35	74.85	84.55	70.26	74.13	82.13
SFIAD Del SFFI And AAML	68.59	74.18	82.44	60.39	73.21	81.80	63.54	73.65	82.05

"Del" denotes removal

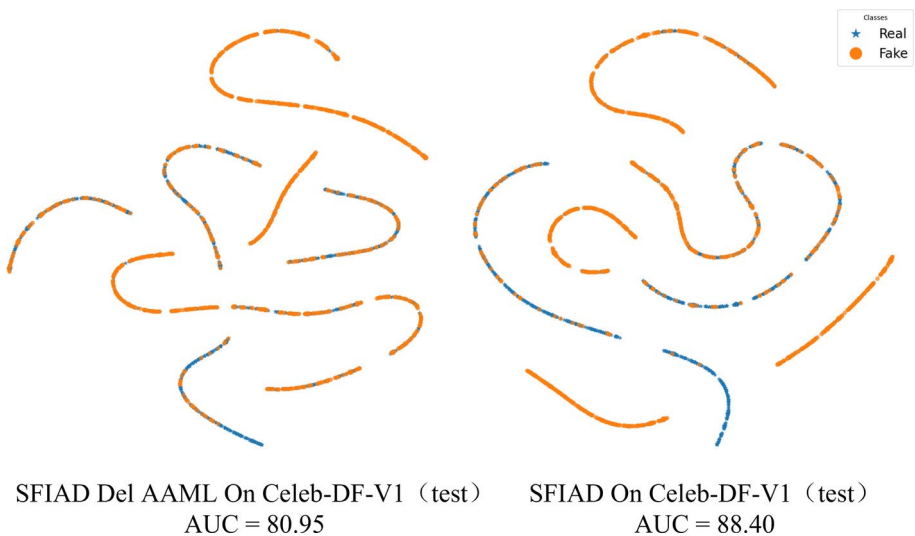
Table 6 To validate the effectiveness of AAML, it was integrated with other methods to assess its impact

Methods	Celeb-DF-V1	Celeb-DF-v2	DFDCP
MesoNet	70.77 ↑	59.83 ↑	58.41 ↓
FWA	69.53 ↑	62.29 ↑	63.10 ↑
Xception	81.56 ↑	76.45 ↑	74.82 ↑
F ³ -Net	64.05 ↑	65.44 ↑	77.26 ↑
RFM	84.31 ↑	77.86 ↑	72.79 ↑
SRM	61.38 ↑	68.27 ↑	74.21 ↑
CORE	65.59 ↑	65.70 ↑	75.68 ↑
Recce	64.72 ↑	66.68 ↑	75.83 ↑
LSDA	84.68 ↑	80.63 ↑	81.39 ↑

The displayed metric is AUC

Table 7 Comparison of AUC metrics of different loss functions combined with Xception on Celeb-DF-V1, Celeb-DF-v2, and DFDCP datasets

Methods	Celeb-DF-V1	Celeb-DF-v2	DFDCP
Xception + Cross-entropy	0.7418	0.7321	0.7365
Xception + PaCoLoss	0.5	0.5	0.5
Xception + AM-Softmax	0.7502	0.7504	0.7422
Xception + AAML	0.8156	0.7645	0.7482

**Fig. 9** Classification boundary comparison of the SFIAD method using t-SNE visualization on the Celeb-DF-V1 dataset

tions and clearer classification boundaries, with an AUC of 88.40. In contrast, after removing AAML (SFIAD Del AAML On Celeb-DF-V1 (test)), the feature distributions became more mixed, the classification boundaries were blurred, and the AUC dropped to 80.95. These experimental results visually demonstrate that the AAML module effectively optimized the classification boundaries, enhancing the model's ability to distinguish between real and forged images, thereby improving the performance of deepfake detection.

Table 8 Impact of Data augmentation on model performance

The displayed metric is AUC

Methods	Celeb-DF-V1	Celeb-DF-v2	DFDCP
SFIAD - Data aug	80.15	72.96	73.40
SFIAD	88.40	79.00	75.88

4.4 Data augmentation impact analysis

To investigate the impact of data augmentation on model performance, we compared the performance of the SFIAD model without data augmentation (SFIAD - Data aug) and the SFIAD model with data augmentation on the Celeb-DF-V1, Celeb-DF-V2, and DFDCP datasets. As shown in Table 8, data augmentation significantly improved the model's performance: on Celeb-DF-V1, the AUC increased from 80.15% to 88.40%; on Celeb-DF-V2, the AUC increased from 72.96% to 79.00%; and on DFDCP, the AUC increased from 73.40% to 75.88%. These experimental results validate the effectiveness of the data augmentation strategy.

5 Conclusion

This study proposes a frame-level deepfake detection framework that combines Spatial-Frequency Feature Integration (SFFI) and Authenticity-Aware Margin Loss (AAML), significantly improving the accuracy and robustness of fake content detection. Our method effectively integrates spatial and frequency domain features of images, enabling the model to more effectively capture subtle differences in forged images. AAML alleviates the class imbalance issue by dynamically adjusting the margins, particularly improving classification performance in cases where authentic samples are scarce. Experimental results on multiple challenging datasets demonstrate that our method outperforms eight existing mainstream detection methods in terms of accuracy, AUC, and AP, confirming its strong potential in deepfake detection. In the future, we plan to extend this framework to video-level deepfake detection and further incorporate spatiotemporal inconsistencies to enhance the model's generalization ability.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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